L.E.College, Morbi

Electrical Engineering Department

Problem Based Learning Activity

Subject Name with Code: Analog and Digital Electronics (BE03009011)

Sr No	Name of Activity	No. of Hours Utilization	Total Hrs	Evaluation criteria
1	Study and simulate logic gates: AND, OR, NOT, NAND, NOR, XOR using Vlab	Duration of task = 2h Preparation of Report = 2h	04	Based on the understanding and applying of Vlab for realisation.
2	Interpretation of Datasheet of IC 741 and Analyze datasheets	Preparing report = 04 Hrs	04	Based on the understanding
3	Technical Video based learning related to the analog and Digital electronics	Duration of video = 5h Report preparation = 5h Total = 10h	10	Report /presentation based on the video learning outcomes.
4	Poster/chart/power point preparation on ADE	Duration = 6 h	06	Based on poster/chart preparation and presentation skills
5	Simulation of all Linear and Non Linear application of op-amp	Use of MATLAB/Multisim for circuit simulation = 06 Hrs	06	Based on the type of application and explanation
6	Industry Visit and Report Preparation	Attend an industry visit (e.g., to a substation, manufacturing unit, renewable energy plant) and prepare a detailed report.05 Hrs	05	Based on the report
7	Mini project: Design and implementation of any application of op-amp	Undertake a small homebased or simulation-based project (10 h)	10	Based on the understanding, implementation and results.